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Codio - Advanced Image Manipulations

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SANDIPAN-DEY

imageGen.py

1# WRITE YOUR CODE HERE

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Guide

Collapse

Image Composition

We learned how to generate images with DALL-E 2 API and manipulate basic properties like size, aspect ratio, brightness, contrast, saturation, and hue. In this lesson, we will explore more advanced image manipulation techniques to create even more variations and artistic effects.

Before that we are going to start with the same steps which are importing our libraries and having our function that generates our prompts.

```
import os
import openai
import secret
from PIL import Image
from io import BytesIO
import requests

openai.api_key = secret.api_key

# Generate the base image
def generate_base_image(prompt):
    response = openai.Image.create(
        prompt=prompt,
        n=1,
        size="512x512"
    )
    return response['data'][0]['url']
```

TRY IT!

Command was successfully executed.

Image setting up

One technique to create more complex images is by composing multiple images together. In this example, we will overlay a generated image of a moon on top of an image of a night sky.

First step we are going to get our function to generate the images and save them into new files.

```
# Generate moon and night sky images
moon_image_url = generate_base_image('moon')
night_sky_image_url = generate_base_image('night sky with stars')

img_data = requests.get(moon_image_url).content
with open('moon_image.jpg', 'wb') as handler:
    handler.write(img_data)

img_data = requests.get(night_sky_image_url).content
with open('night_sky_image.jpg', 'wb') as handler:
    handler.write(img_data)
```

TRY IT!

Command was successfully executed.

Click here to refresh your moon image

Click here to refresh your night sky image

If you don't like the pictures generated, generate them again. If you like one and not the other comment out the code generation for the one you want. If you are happy with both, please comment out the generation code so we don't have to waste time generating images again and saving them over the image you want. At the end you should have something more or less like this :

```
"""
# Generate moon and night sky images
moon_image_url = generate_base_image('moon')
night_sky_image_url = generate_base_image('night sky with stars')

img_data = requests.get(moon_image_url).content
with open('moon_image.jpg', 'wb') as handler:
    handler.write(img_data)

img_data = requests.get(night_sky_image_url).content
with open('night_sky_image.jpg', 'wb') as handler:
    handler.write(img_data)
"""
```

Now we are ready to combine our images.

What is the purpose of using the `requests.get()` method in this lesson?

☐ To import the necessary libraries

☐ To create the image generation function

☒ To download the generated image from the URL

☐ To combine the moon and night sky images

The `requests.get()` method is used to download the generated image from the URL returned by the `generate_base_image` function. The image data is then saved into a file using a with `open()` statement.

Check It!

Next

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https://codio.com/sandipan-dey/advanced-image-manipulations